Solution

Let $(a,b) \in \mathbb{R}^2$ be an arbitrary point. We shall investigate continuity of f at (a,b).

That is, we shall check if all the three conditions for continuity hold for f at (a,b).

To check first condition, note that $f(a,b) = \dfrac{3a-5b+8}{a^2+b^2+1}$ is defined.

Next we want to find if $\lim\limits_{(x,y)\to(a,b)} f(x,y)$ exists or not.

So we calculate $\lim\limits_{(x,y)\to(a,b)} (3x-5y+8) = 3a-5b+8$ and $\lim\limits_{(x,y)\to(a,b)} (x^2+y^2+1) = a^2+b^2+1 \neq 0$.

Thus, by the properties of limits, we see that

$$\lim_{(x,y)\to(a,b)} f(x,y) = \frac{\lim\limits_{(x,y)\to(a,b)} (3x-5y+8)}{\lim\limits_{(x,y)\to(a,b)} (x^2+y^2+1)} = \frac{3a-5b+8}{a^2+b^2+1} = f(a,b) \text{ = L exists.}$$

Now we note that $\lim\limits_{x,y\to(a,b)} f(x,y) = L = f(a,b)$. Hence f satisfies all the three conditions for continuity of f at (a,b). Since (a,b) is an arbitrary point in $\mathbb{R}^2$, we conclude that f is continuous at every point of $\mathbb{R}^2$. ■

Example 8.9

Consider $f(x,y) = \dfrac{xy}{x^2+y^2}$ if $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Show that f is not continuous at $(0,0)$ and continuous at all other points of $\mathbb{R}^2$.

Solution

Note that f is defined for every $(x,y) \in \mathbb{R}^2$. First let us check the continuity at $(a,b) \neq (0,0)$. Let us say, just for instance, $(a,b) = (2,5)$. Then $f(2,5) = \dfrac{10}{29}$. Then, as in the above example, we calculate $\lim\limits_{(x,y)\to(2.5)} xy = 2(5) = 10$ and $\lim\limits_{(x,y)\to(2.5)} x^2+y^2 = 2^2+5^2 = 29 \neq 0$.

$$\text{Hence } \lim_{(x,y)\to(2,5)} \frac{xy}{x^2+y^2} = \frac{10}{29}.$$

Since $f(2,5) = \dfrac{10}{29} = \lim\limits_{(x,y)\to(2,5)} \dfrac{xy}{x^2+y^2}$, it follows that f is continuous at $(2,5)$.

Exactly by similar arguments we can show that f is continuous at every point $(a,b) \neq (0,0)$. Now let us check the continuity at $(0,0)$. Note that $f(0,0) = 0$ by definition. Next we want to find if $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$ exists or not.

First let us check the limit along the straight lines $y = mx$, passing through $(0,0)$.

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{mx^2}{(1+m^2)x^2} = \frac{m}{1+m^2} \neq f(0,0), \text{ if } m \neq 0.$$

So for different values of m, we get different values $\dfrac{m}{1+m^2}$ and hence we conclude that $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$ **does not exist**. Hence f cannot be continuous at $(0,0)$. ■

Chapter 8 Differentials and Partial Derivatives.indd 72 26-02-2020 17:01:48